THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The altitude on the hypotenuse of the right triangle with legs 7 and 24 is of length (cm): 1

(a) $\frac{168}{25}$ **(b)** $\frac{31}{2}$ **(c)** $\frac{25}{2}$ **(d)** 84

2. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 100$ and $ar(PQR) = 256$ and $AB = 10$, then $PQ =$ 1

(a) 10 **(b)** 21 **(c)** 128 **(d)** 16

3. In $\triangle ABC$ right-angled at B, $AB = 6$, $BC = 8$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{32}{5}$ **(b)** $\frac{24}{5}$ **(c)** $\frac{18}{5}$ **(d)** 3

4. A square has diagonal $11\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 121 **(b)** 22 **(c)** 484 **(d)** 242

5. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 5$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 4: 1

(a) 45 **(b)** 53 **(c)** 25 **(d)** 49

6. In a right triangle with hypotenuse $c = 29$ and one leg $a = 20$, the area (sq. units) is: 1

(a) 210 **(b)** $\frac{609}{2}$ **(c)** 420 **(d)** 290

7. The altitude on the hypotenuse of the right triangle with legs 20 and 21 is of length (cm): 1

(a) $\frac{41}{2}$ **(b)** $\frac{29}{2}$ **(c)** $\frac{420}{29}$ **(d)** 210

8. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 54$ and $ar(PQR) = 600$ and $AB = 6$, then $PQ =$ 1

(a) 200 **(b)** 23 **(c)** 6 **(d)** 20

9. **Assertion (A):** Any two equilateral triangles are similar. 1

Reason (R): If two angles of one triangle are equal to two angles of another triangle, the triangles are similar.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** If two similar triangles have areas $16\ \text{cm}^2$ and $81\ \text{cm}^2$, their corresponding sides are in the ratio $4 : 9$. 1

Reason (R): The ratio of the areas of two similar triangles equals the ratio of their corresponding sides.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the altitude on the hypotenuse of the right triangle with legs 8 cm and 15 cm. **2**

12. In $\triangle ABC$, $DE \parallel BC$, $AD = 5$ cm, $AB = 8$ cm and $BC = 32$ cm. Find DE. **2**

13. The areas of two similar triangles are 50 cm² and 72 cm². If a side of the first is 5 cm, find the corresponding side of the second. **2**

14. Find the altitude on the hypotenuse of the right triangle with legs 7 cm and 24 cm. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 18$ cm and $DC = 24$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. **3**

16. Find the altitude of an equilateral triangle of side 14 cm, in terms of $\sqrt{3}$. **3**

17. $\triangle ABC$ is right-angled at B with $AB = 20$ cm and $BC = 21$ cm; $BD \perp AC$. Find AD, DC and BD. **3**

OR

In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 8$ cm and $DC = 18$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. **3**

18. Find the altitude of an equilateral triangle of side 18 cm, in terms of $\sqrt{3}$. **3**

OR

$\triangle ABC$ is right-angled at B with $AB = 9$ cm and $BC = 40$ cm; $BD \perp AC$. Find AD, DC and BD. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A 17 m long ladder rests against a vertical wall with its top at a window 15 m above the ground. **5**

(i) Which side of the right triangle formed is the hypotenuse?

(ii) Find the distance of the foot of the ladder from the wall.

(iii) Find the perimeter of the triangle formed by the ladder, the wall and the ground.

(iv) The top of the ladder slides down the wall to a height of 8 m. How far is the foot of the ladder from the wall now, and how far did the foot move?

20. Two poles of heights 19 m and 7 m stand on level ground 9 m apart. Find the distance between their tops. **5**

OR

A girl 2 m tall walks away from a lamp post 8 m high at 3 m/s. Find the length of her shadow after 4 seconds. **5**

END OF QUESTION PAPER — Total: 40 Marks

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